



Research



Cite this article: Caillet J, Seroussi H, Nowicki S. 2026 Uncertain ground: impact of bed topography on Antarctic Ice Sheet projections. *Phil. Trans. R. Soc. A* **384**: 20240543.

<https://doi.org/10.1098/rsta.2024.0543>

Received: 7 July 2025

Accepted: 6 October 2025

One contribution of 16 to a Theo Murphy meeting issue ‘Next generation ice-sheet bed measurements’.

Subject Areas:

glaciology

Keywords:

ice sheet modelling, ice sheet dynamics, Antarctica, bedrock, uncertainty

Author for correspondence:

Helene Seroussi

e-mail: helene.l.seroussi@dartmouth.edu

[†] Joint first authors.

Electronic supplementary material is available online at <https://doi.org/10.6084/m9.figshare.c.8405335>.

Uncertain ground: impact of bed topography on Antarctic Ice Sheet projections

Justine Caillet^{1,†}, Helene Seroussi^{1,†} and Sophie Nowicki²

¹Thayer School of Engineering, Dartmouth College, Hanover, NH, USA

²University at Buffalo, Buffalo, NY, USA

JC, 0000-0001-9435-7585; HS, 0000-0001-9201-1644; SN, 0000-0001-6328-5590

Despite decades of effort to map bed elevation under the Antarctic Ice Sheet, significant gaps remain in this geometric boundary, controlling ice sheet flow and grounding line dynamics. The impact of such uncertainties on ice flow simulations of the Antarctic evolution, however, has received little attention. Here, we investigate their impact on the Antarctic evolution at the continental scale as well as the regional scale for the Bellingshausen and Aurora basins and compare them to the impact of climate forcing scenarios. Using error estimates reported in BedMachine Antarctica, our simulations show that bed topography affects the Antarctic contribution to sea level by more than 40 cm in 2150 and 1 m by 2300, comparable to changes caused by different emission scenarios. Variations in grounding line retreat and mass loss are especially important in the Amundsen Sea, Ross and Filchner–Ronne basins. The impact of bedrock uncertainties is even larger on regional- and glacier-scale simulations with a higher spatial resolution, resulting in more variations in grounding line positions and mass change. Overall, these results suggest that errors in bedrock elevation under the Antarctic Ice Sheet are a critical but underexplored source of uncertainty. Additional observations in critical regions are needed to help reduce these systemic uncertainties.

This article is part of the Theo Murphy meeting issue ‘Next generation ice-sheet bed measurements’.

1. Introduction

Simulations of the Antarctic Ice Sheet show significant ice mass loss over the next centuries, extending the trend observed over the past few decades [1,2]. Ensembles of Antarctic model simulations performed as part of the Ice Sheet Model Intercomparison for CMIP6 (ISMIP6) [3,4] suggest up to 0.5 m of sea-level equivalent (SLE) by 2100 [5,6] and up to 6.9 m SLE by 2300 [7] under high greenhouse gas emission scenarios. Models agree that most of the mass loss comes from the acceleration and thinning of fast ice streams in large marine basins of West Antarctica, as well as a few basins in East Antarctica, as ice shelves thin and collapse in response to warmer ocean waters reaching ice shelf cavities [8,9].

While models generally agree on the regions contributing to Antarctic mass loss and the main processes driving it, significant uncertainties remain in the projected magnitude and timing of this mass loss [7,10]. These uncertainties come from multiple sources, including not only differences in greenhouse gas emission scenarios but also the range of oceanic and atmospheric conditions simulated by climate models under similar scenarios. Additional sources of uncertainty include the physical processes included in ice flow models, their parameterizations, the methods used for model initialization and the spatial resolution of the simulations [6,11,12].

Analysis of ensembles of Antarctic Ice Sheet simulations, such as those performed as part of SeaRISE [13,14] and ISMIP6 [4,5,11], provides information about uncertainties for different climate scenarios. The ISMIP6 ensemble of Antarctic experiments suggests that under high-emission scenarios, the Antarctic Ice Sheet could lose between -8 and 50 cm SLE by 2100 [5,6]. Additional simulations performed until 2300 show that such mass loss would increase to 6.9 m SLE by 2300 [7]. Further analysis highlights the respective contributions of ice flow model choice, climate forcing and their interactions to the overall uncertainty in these projections [6]. It suggests that by 2100 and for the RCP8.5 and SSP5-8.5 high-emission scenario, more than half of the uncertainty comes from the choice of ice flow models (including all model differences), while the rest of the uncertainty is equally split between the choice of climate model, the interaction between ice and climate and the uncertainty introduced by emulating missing results. The exact partitioning between these different terms varies over time and for different basins, but overall, the choice of ice flow model remains the major source of uncertainty for the 2015–2300 period simulated as part of ISMIP6 Antarctica 2100 and Antarctica 2300 [6,7].

Many previous studies performed with single ice flow models have also investigated uncertainties coming from a number of ice sheet physical processes and model parameters with continental-scale simulations of the Antarctic Ice Sheet. Studies investigating the impact of basal sliding and subglacial hydrology show that including a representation of subglacial water processes increases Antarctic ice mass loss, as do sliding laws with higher exponents [15,16]. Idealized experiments have investigated the impact of basal thawing of previously frozen areas at the ice base by changing basal friction from values characteristic of frozen beds to those typical of thawed beds [17]. These experiments lead to a rapid acceleration of mass loss, especially in George V-Adélie Land and Enderby–Kemp regions in East Antarctica. Studies investigating the role of initial conditions and the historical trajectory of the simulations highlight that models with relatively similar initial configurations may lead to significantly varying future evolution, with mass loss varying by up to 50% [11,18]. Choices made at the ice–ocean interface to convert oceanic properties into ice shelf basal melt create large variations in the ice shelf geometry and therefore the dynamic response of Antarctic ice streams: comparison of simulations performed with different ice shelf melt parameterizations and their calibration suggests three to four times more sea-level contribution for the most sensitive parameterizations compared to the least sensitive ones [19]. Choices made at the ice–atmosphere interface also result in large discrepancies after 2100 under high-emission scenarios, as melt-elevation feedback becomes an amplifying driver of mass loss when the increase in surface runoff outweighs the increase in snow accumulation. Oceanic and atmospheric drivers combine and result in a complete collapse of West Antarctica and a significant retreat of the grounding line in East Antarctica [20]. On the other hand, sensitivity studies

carried out with different approximations of the stress balance equations show relatively limited impact on the Antarctic mass loss on century time scales [19].

Regional studies have also investigated the role of different ice rheology laws [21] and friction laws [22] to better capture complex spatial patterns of observed velocities and the impact of calving [23–25] on grounding line retreat, melt rate and mass loss for Pine Island and Thwaites glaciers in the Amundsen Sea sector. In addition, ensembles of simulations can help evaluate the role of internal climate variability by introducing stochastic variability in climate forcing. Such ensembles, also performed on Thwaites Glacier, indicate that instabilities causing rapid ice sheet retreat are amplified by the uncertainty introduced by variability in climate forcing, causing rapid uncertainty growth during such retreat phases [26].

One source of uncertainty that has received less attention is the impact of bedrock elevation. Despite the diversity of ice flow models, processes and parameterizations used in Antarctic projections, all simulations rely on a limited number of bedrock elevation datasets, reflecting the scarcity of such observational data. Out of the 17 models that took part in ISMIP6 Antarctica 2300 [7], 13 used BedMachine Antarctica [27] while four relied on Bedmap [28], making it impossible to investigate the role of using different bed topography. The large increase in the number of observations of Antarctic bedrock topography and the improvement of algorithms to map it have greatly improved the quality of bed products and reduced the associated uncertainty over the past decade. However, substantial uncertainties persist in many less-explored regions, where reported errors can still reach hundreds of metres [27,29].

Studies on idealized configurations based on flow line models have shown that bedrock topography should be described at a resolution of approximately 1 km [30]. For Thwaites Glacier, current uncertainties in bedrock elevation alone can lead to a 22 cm variation in its sea-level contribution by 2200 [31]. Reducing this uncertainty to under 2 cm would require bedrock measurements with a horizontal resolution of 2 km and a vertical accuracy better than 8 m. However, the influence of bedrock uncertainty on the future evolution of the entire Antarctic Ice Sheet remains largely unknown. In this study, we investigate the impact of current uncertainties in bed topography [27,29] on the future evolution of the Antarctic Ice Sheet and its contribution to sea-level rise.

Most Antarctic and Greenland ice flow simulations used for sea-level projections in previous Intergovernmental Panel on Climate Change reports are based on continental-scale simulations. However, recent results suggest that simulations at the regional or glacier scale better capture recent changes and lead to faster ice sheet response [32]. These results have been performed for the Greenland Ice Sheet, but no similar experiments have been applied to Antarctic glaciers. We therefore perform simulations at a continental scale, as well as at a regional scale for the Belling-shausen and Aurora basins, and at a glacier scale for all the glaciers in these two regions, and compare the results for these three scales for the two regions selected, including the role of bed uncertainty.

2. Methods

(a) Ice flow model

We use the Ice sheet and Sea-level System Model (ISSM) [33] for all ice sheet simulations. The overall model set-up and initialization are similar to the procedure used for the ISMIP6 JPL-ISSM [5] and DC-ISSM [7] models, and we summarize here the main model characteristics. The initial geometry is interpolated from BedMachine Antarctica v.4 for bedrock elevation, surface elevation and ice front positions [27,29]. The grounding line position is determined by the flotation condition, with sub-element friction parameterization [34] and sub-element ice shelf melt parameterization [35] applied on partially grounded and partially floating mesh elements. The stress balance equation is solved using the Shelfy-Stream Approximation [36] with a rheology based on Glen's flow law [37]. Basal sliding is simulated using the Budd sliding law and assuming a

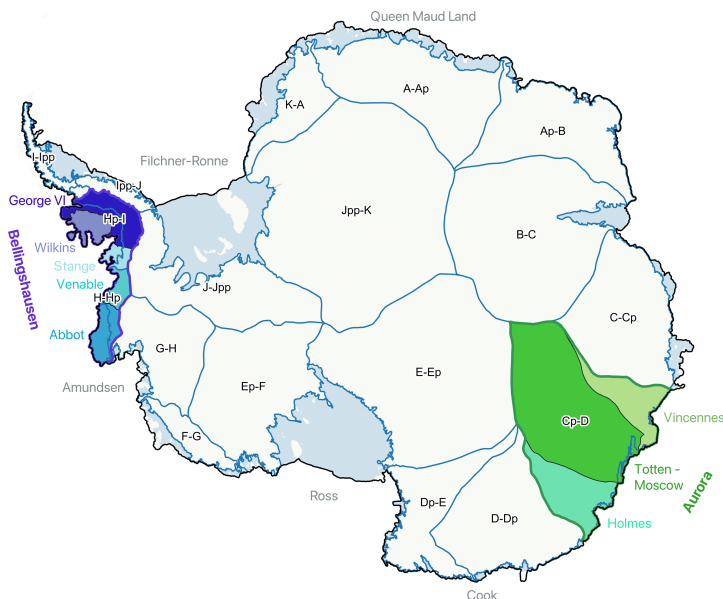


Figure 1. Overview figure of the Antarctic Ice Sheet, with regional-scale Aurora (Cp-D, IMBIE basin definition [44]) and Bellinghshausen (Hp-I and H-Hp) sectors simulated. Glacier-scale domains simulated include: Vincennes, Totten–Moscow and Holmes for the Aurora region, and George VI, Wilkins, Stange, Venable and Abbot for the Bellinghshausen region.

subglacial hydrologic system perfectly connected to the ocean to evaluate the effective pressure [38].

The basal friction over the grounded ice and the rheology over the ice shelves are determined using data assimilation [39] to match observed surface velocity [40], while the ice rheology over the grounded ice is estimated using a thermal model and assuming thermal steady-state conditions [41]. Sub-ice shelf melt is based on the ISMIP6 quadratic non-local basal melt parameterization and calibration, as described in the ocean protocol [42]. The ice front position is tracked using the implicit level set method [43]. It is allowed to retreat only when the ice thins below 10 m and remains fixed otherwise.

(b) Continental-, regional- and glacier-scale simulations

We perform experiments at continental, regional and glacier scales to compare the impact of model resolution and calibration at different scales, as calibration at regional to glacier scale is suggested to provide a more accurate calibration of smaller glaciers than a continent-wide calibration [32].

Continental-scale simulations include the entire Antarctic Ice Sheet (figure 1) and have a resolution varying from 3 to 60 km, with refinement in the shear margins as well as in areas around the grounding line and where the grounding line is expected to retreat (table 1). These simulations have a two-dimensional mesh of approximately 210 000 elements. Following initialization, the model is spun up for 6 years between 2002 and 2007 to allow some relaxation of the model geometry, grounding lines and velocity, and to limit unphysical changes caused by inconsistent datasets at the beginning of the experiments [45]. The 1995–2017 ocean thermal forcing climatology from ISMIP6 [42] and the 2002–2007 average surface mass balance of the RACMO v2.3 regional climate model [46] are used as climate conditions during this relaxation.

Regional simulations for the Bellinghshausen and Aurora basins include these entire basins, from the ice divide to the coast, using the extent of the IMBIE basins [44]. The resolution is increased compared to continental-scale simulations to reach 1 km along the coast and up to 50

Table 1. List of experiments. *Standard bed* refers to the BedMachine Antarctica v.4 bed topography. *Max bed* and *min bed* correspond to topographies obtained by adding and subtracting, respectively, the uncertainty provided in BedMachine Antarctica v.4 from the standard bed.

domain	bed topography	scenario	minimum resolution	number of elements
Antarctic Ice Sheet	standard/max/min	SSP5-8.5	3 km	209 770
Antarctic Ice Sheet	standard/max/min	SSP1-2.6	3 km	209 770
Bellingshausen region	standard/max/min	SSP5-8.5	1 km	154 697
Bellingshausen region	standard	SSP1-2.6	1 km	154 697
Aurora region	standard/max/min	SSP5-8.5	1 km	88 139
Aurora region	standard	SSP1-2.6	1 km	88 139
Vincennes glacier	standard/max/min	SSP5-8.5	500 m	68 114
Vincennes glacier	standard	SSP1-2.6	500 m	68 114
Totten–Moscow glacier	standard/max/min	SSP5-8.5	500 m	89 143
Totten–Moscow glacier	standard	SSP1-2.6	500 m	89 143
Holmes glacier	standard/max/min	SSP5-8.5	500 m	69 170
Holmes glacier	standard	SSP1-2.6	500 m	69 170
George VI glacier	standard/max/min	SSP5-8.5	500 m	129 036
George VI glacier	standard	SSP1-2.6	500 m	129 036
Wilkins glacier	standard/max/min	SSP5-8.5	500 m	91 555
Wilkins glacier	standard	SSP1-2.6	500 m	91 555
Stange glacier	standard/max/min	SSP5-8.5	500 m	44 003
Stange glacier	standard	SSP1-2.6	500 m	44 003
Venable glacier	standard/max/min	SSP5-8.5	500 m	47 702
Venable glacier	standard	SSP1-2.6	500 m	47 702
Abbot glacier	standard/max/min	SSP5-8.5	500 m	96 447
Abbot glacier	standard	SSP1-2.6	500 m	96 447

km in slow-flowing areas. The regional models have approximately 155 000 and approximately 88 000 elements for the Bellingshausen and Aurora regions, respectively. Figure 1 shows the extent of model domains for these two regions at the beginning of the experiments. The boundary conditions used at the divide of the sectors include imposed velocity and thickness from the continental-scale simulations at the end of the spin-up in order to use conditions most similar to the Antarctic-scale experiments. These two regions were selected as they both show interesting changes by 2300 in ISMIP6 experiments [7] and have been less investigated than regions like the Amundsen Sea sector. The Bellingshausen region displays relatively limited error in bedrock topography, with errors below 50 m in most areas and locally reaching 100 m (figure 2b). The Aurora region, on the other hand, displays much larger errors that are often above 200 m and can reach up to 600 m. These qualitative errors represent uncertainty from the different methods that are used in BedMachine Antarctica [29], including errors from direct bed observations from radar sounding, a quantitative assessment of error based on the mass conservation in fast-flowing areas [47] and an heuristic error from the ice flow perturbation analysis in slow-flowing regions [48].

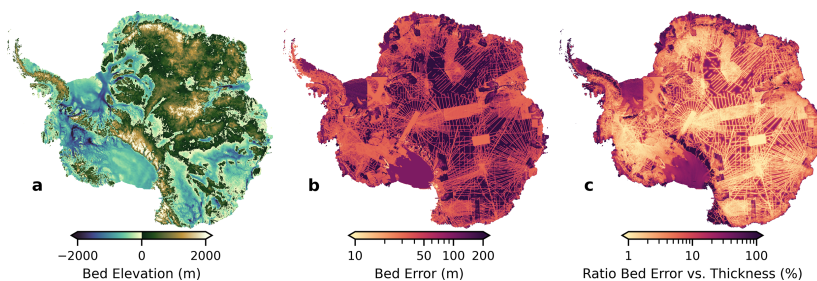


Figure 2. Bedrock topography under the Antarctic Ice Sheet from BedMachine Antarctica v.4 [29] (a, in m), uncertainty in bedrock reported as absolute error (b, in m) and relative error to the ice thickness (c, in % of the ice thickness).

The two regions selected also show contrasting topography, with wide glaciers flowing over deep bedrock mostly below sea level in the Aurora region (figure 2a), while the Bellingshausen region is a very mountainous region with steep topographic changes.

Glacier-scale simulations are also performed for these two regions. The Aurora region is divided into three basins, Vincennes, Totten–Moscow and Holmes, while the Bellingshausen region is divided into five basins, George VI, Wilkins, Stange, Venable and Abbot (figure 1). The extent of each glacier-scale simulation is based on the basin divides [49], and the combined extent of the glacier-scale domains matches that of the regional-scale domains. Boundary conditions are drawn from the continent-wide Antarctic simulations, similar to the regional-scale simulations. The spatial resolution of these simulations varies between 500 m and 50 km, leading to models with approximately 44 000 to approximately 129 000 elements (table 1).

For the Aurora and Bellingshausen regions, we compare results extracted from continental-scale simulations with those from the regional-scale simulations and the glacier-scale simulations, in which results from all the glaciers are added together.

(c) Experiments

Following the relaxation, a historical run is performed for the continental-, regional- and glacier-scale configurations between 2007 and 2015 using 2007–2015 annual surface mass balance time series from RACMO v2.3 [46] and 1995–2017 oceanic thermal forcing climatology from ISMIP6 [42].

All projection experiments start in 2015 and run until 2300, following the ISMIP6 Antarctic framework [4,5] and its extensions until 2300 [7]. Experiments are performed for a low-emission scenario, SSP1-2.6, and for a high-emission scenario, SSP5-8.5. Table 1 lists a summary of all the experiments performed and their main characteristics. Oceanic and atmospheric conditions for these two scenarios are extracted from the UK Earth System Model [50], as this model was used for both scenarios in previous ISMIP6 simulations [7].

Experiments are run at continental-, regional- and glacier-scale for both scenarios with the standard bedrock geometry of BedMachine Antarctica v.4, referred to as standard bed. In addition, experiments are also performed with lower and higher bed topography (table 1). The additional bedrock elevations are derived by adding or subtracting the bed uncertainty listed in BedMachine Antarctica, which is suggested to capture a two-sigma error [29], and are referred to as max bed and min bed, respectively. For these experiments with different bedrock topography, the rest of the geometry, including ice front and surface elevation, is kept similar to the standard case, while ice thickness and grounding line position are adjusted based on the flotation criterion. The entire model initialization, including the friction and rheology inversions, is performed independently for the three bed topographies, but starting with the basal friction and rheology fields from the standard bed case as an initial guess for the inversions, to ensure models are as similar as possible and limit differences introduced by the inversions.

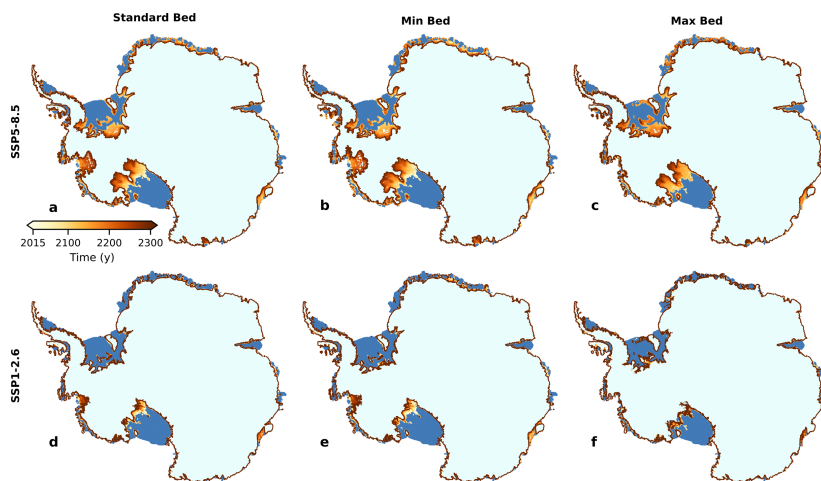


Figure 3. Grounding line evolution between 2015 and 2300 (every 10 years) for continental-scale simulations for the high (SSP5-8.5, a, b and c) and low (SSP1-2.6, d, e and f) emission scenarios and the standard bedrock (standard bed, a and d), lower bedrock (min bed, b and e) and higher bedrock (max bed, c and f) topographies. The background map shows the grounded areas (light blue) and floating areas (dark blue) in the year 2015.

3. Results

(a) Continental scale

The continental-scale evolution of the Antarctic Ice Sheet is similar to that described previously in ISMIP6 Antarctica 2300 when the standard bed is used. Simulations indicate large grounding line retreats for the high-emission scenario in West Antarctica, affecting Ross and Ronne–Filchner ice shelves as well as the Amundsen Sea sector. More modest retreats are observed in East Antarctica, affecting Totten, Moscow University and Cook ice shelves as well as many glaciers in Queen Maud Land (figure 3a). In contrast, the low-emission scenario suggests relatively limited changes, with slow retreats of Ross ice shelf as well as Thwaites, Totten and Moscow University glaciers (figure 3d).

Simulations based on different bed topographies exhibit retreat in the same areas overall, but with varying rates and timing of retreat. The higher bedrock topography, max bed, leads to the largest difference in grounding line retreat compared to the standard bed, with no retreat in the Amundsen Sea sector for both emission scenarios (figure 3c,f). In the high-emission scenario, minor retreat is simulated in the sector of Cook ice shelf. The grounding line retreat of Ross ice shelf is also delayed, leading to less retreat overall by 2300, and no retreat is simulated in this region in the low-emission scenario. The lower bedrock topography causes less difference with the standard bed, and the results of these two simulations more closely align. Small differences in the retreat rates can be noted, for example, with enhanced retreat of Pine Island glacier, in the Siple Coast, or for Cook, Abbot and Getz ice shelves for the high-emission scenario (figure 3b). Results do not suggest noticeable differences for the low-emission scenario (figure 3e).

The ice front retreat (electronic supplementary material, fig. S1), which happens when the ice thickness becomes thinner than 10 m, affects all floating ice shelves around Antarctica in the high-emission scenario and is caused by the large increase in sub-ice shelf melt, from approximately 1000 Gt yr^{-1} at the beginning of the simulations to more than $10\,000 \text{ Gt yr}^{-1}$ during the second half of these simulations (figure 4b). Areas initially floating, as well as those that become afloat owing to grounding line retreat, experience large retreats (electronic supplementary material, fig. S1). Since ice front retreat is primarily driven by ocean-induced melt rates, the overall retreats are very similar across all three bed topographies, with differing ice front retreat only observed

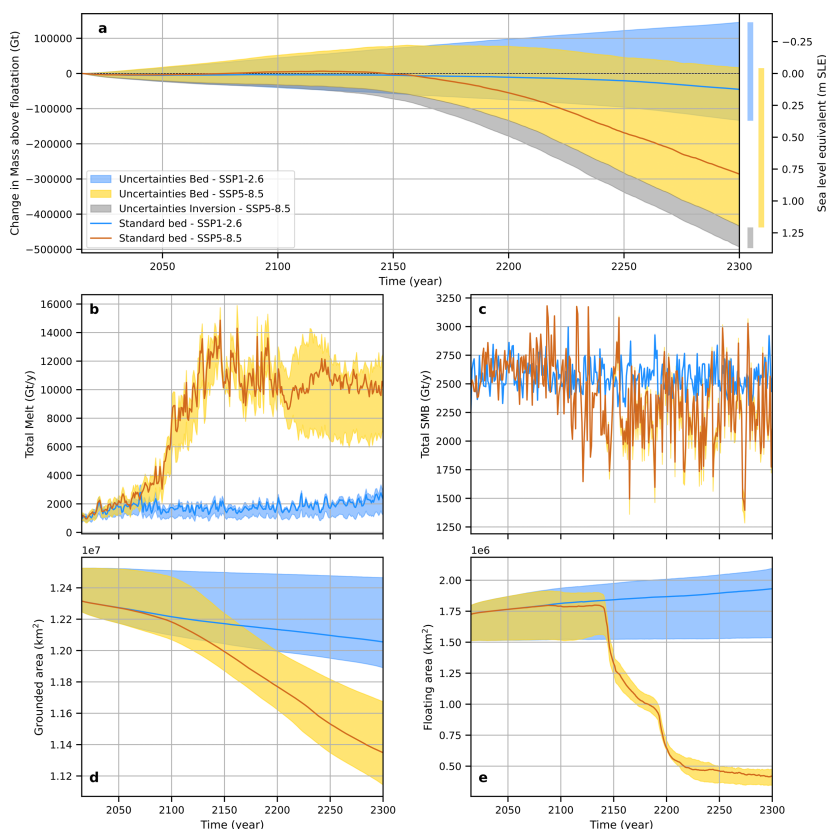


Figure 4. Comparison of the change in mass above flotation and SLE (a), total sub-ice shelf melt (b), total surface mass balance (c), grounded area (d) and floating area (e) for continental-scale simulations. The blue and red lines refer to the low- and high-emission scenarios, respectively, for the standard bed experiments. The light blue and yellow shadings show the range of results for the three different bed experiments (standard bed, min bed and max bed) for the low- and high-emission scenarios, respectively. The grey shading on panel (a) shows the range of results owing to the initialization and inversion procedures carried on the min bed experiment for the high-emission scenario. The right-hand side of panel (a) shows the uncertainty range in 2300 related to bed elevation (blue and yellow for low- and high-emission scenarios, respectively) and initialization procedure (grey).

in regions where grounding line retreats vary. Experiments with low-emission scenario forcings show very limited ice front retreat for all three bed elevations (electronic supplementary material, fig. S1), as ice shelf melt remains below 3000 Gt yr^{-1} (figure 4b).

Figure 4a illustrates that uncertainties in bedrock elevation cause significant changes in the Antarctic contribution to sea level over both short and long time scales. In 2150, mass loss ranges from -0.21 to 0.15 m SLE for both emission scenarios, and by 2300, this range widens to -0.40 to 0.37 m SLE under the low-emission scenario and -0.04 to 1.20 m SLE for the high-emission scenario. Differences in bedrock elevation contribute more to the spread in Antarctic sea-level contribution than the emission scenario itself until approximately 2250.

Since climate forcing is identical for all three bed experiments, the surface mass balance is nearly identical for a given scenario (figure 4c). It is prescribed independently of the evolving ice sheet and is only modified by changes in the ice extent. Ocean conditions are also initially similar, but as ice shelves thin and grounding lines retreat, differences in local ocean forcing and ice shelf extent begin to emerge. Figure 4b shows the evolution of the total ice shelf melt for the continental-scale simulations. The spread in melt rates increases sharply over time, reaching $7100\text{--}12\,700 \text{ Gt yr}^{-1}$ by 2300 under the high-emission scenario and $1200\text{--}3000 \text{ Gt yr}^{-1}$ under the low-emission

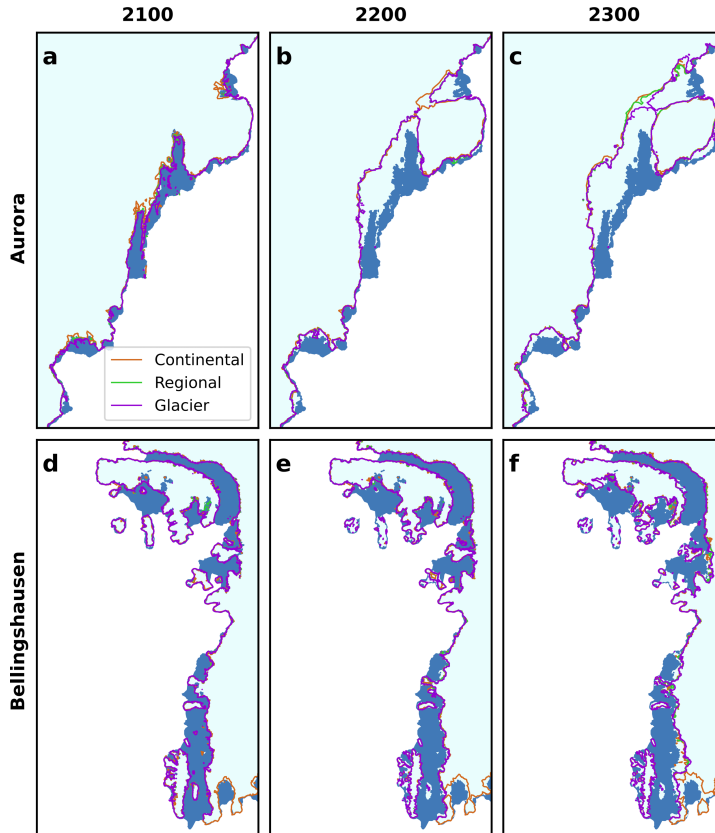


Figure 5. Comparison of grounding line positions for the continental-, regional- and glacier-scale simulations in 2100 (a and d), 2200 (b and e) and 2300 (c and f) for Aurora (a, b and c) and Bellingshausen (d, e and f) regions. This figure shows simulations performed with the standard bed topography and for the high-emission scenario.

scenario. Initial grounded and floating areas vary between 1.23 and 1.25×10^7 km² and 1.51 and 1.80×10^6 km², respectively, reflecting differences in initial grounding line positions owing to bedrock variations (figure 4d,e). These area differences evolve slowly, except for the floating area in the high-emission scenario, which begins to decrease significantly around 2140 owing to rapid ice shelf retreat from enhanced ocean-induced melt.

(b) Regional and glacier scale

Regional- and glacier-scale simulations show results consistent with those of the continental-scale simulations for both regions overall. Grounding line positions using the standard bed (figure 5) show similar patterns for 2100, 2200 and 2300, with only minor differences. One notable exception occurs between Totten–Moscow and Vincennes glaciers after 2200 (figure 5b,c), where the grounding line shows an unrealistically narrow grounded area at the boundary between the two basins. This artefact results from grounding line retreat along the edge between the two glacier-scale domains and the way boundary conditions are handled on this shared boundary (see §4a).

The Aurora region is particularly sensitive to variations in bed topography, which dominate the uncertainty throughout the simulation period (figure 6a). Unlike the Antarctic Ice Sheet as a whole, this region shows a low sensitivity to the emission scenario. The mass loss is relatively limited for the standard bed experiments at all scales and for all scenarios until 2150, after which it

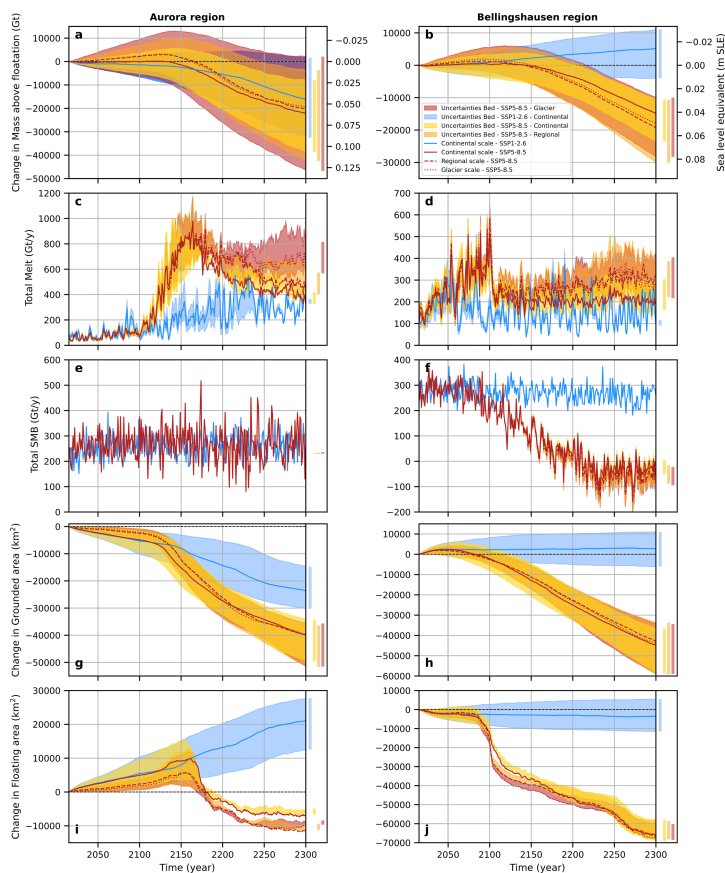


Figure 6. Comparison of continental-, regional- and glacier-scale results for the Aurora (left column) and Bellingshausen (right column) regions showing the change in volume above flotation (a and b), ocean-induced melt (c and d), surface mass balance (e and f), change in grounding area (g and h) and change in floating area (i and j). Results at the glacier scale include values from all glaciers composing a region, while results are extracted for the two regions from the continental-scale simulations to provide comparable numbers. Shaded areas show the spread of results caused by the three bedrock topographies at continental-, regional- or glacier-scale for the high-emission scenario and only at continental scale for the low-emission scenario.

starts increasing to reach approximately 0.05 m SLE in 2300 (figure 6a and electronic supplementary material, fig. S3a). Regional-scale and glacier-scale simulations display less mass loss than continental-scale ones for both scenarios in the standard bed case.

Uncertainty from different bed topography has a greater impact on regional- and glacier-scale simulations compared to continental-scale ones, with mass loss in the Aurora region differing by an additional 0.02 and 0.04 m SLE, respectively, corresponding to a 14 and 30% increase in uncertainty, respectively, by 2300 for the high-emission scenario. The total ice shelf melt in this region increases gradually until 2120 and then more rapidly for the high-emission scenario. It starts decreasing around 2175, corresponding to a reduction in the floating area extent (figure 6g). Both melt rates and uncertainties in melt rates increase with higher resolution as ice shelves start responding differently. Changes in the grounded and floating areas display patterns similar to those described at the continental scale (figure 6g,i).

The sea-level contribution from the Bellingshausen region displays a greater sensitivity to the emission scenario, with the low-emission scenario suggesting a low and gradual mass increase, while the high-emission scenario displays a rapid and increasing mass loss starting around 2150 (figure 6b). Simulations at different scales display very similar grounding line positions and mass loss (figures 5 and 6b). Continental-scale results, however, underestimate mass loss by 0.10 m SLE

for the high-emission scenario by 2300. The impact of bed topography causes a spread of results that is larger at the regional and glacier scale, similar to what is observed for the Aurora region. Differences between the regional and glacier scale are relatively limited (figure 6b and electronic supplementary material, fig. S3b). Mass loss for the high-emission scenario with the different bed topographies varies between 29 and 62 mm SLE for the continental-scale simulations, between 29 and 83 mm SLE for the regional-scale simulations and between 28 and 78 mm SLE for the glacier-scale simulations in 2300.

The ocean-induced ice shelf melting is overall higher in the high-emission scenario compared to the low-emission scenario (figure 6d), but the largest difference between the two scenarios comes from the surface mass balance: it decreases steadily from 300 Gt yr⁻¹ to -50 Gt yr⁻¹ between 2050 and 2225 for low emissions (figure 6f). The grounded and floating areas are similar in all simulations for a given scenario (figure 6h,j). They both decrease faster for the high-emission scenario, with similar results for the different scales. The uncertainty owing to bed topography increases over time for the grounded area, while it remains mostly constant for the floating area as the overall size of the ice shelves decreases owing to increased and sustained ice shelf melting.

4. Discussion

(a) Continental-, regional- and glacier-scale simulations

Our results provide insights into how uncertainties in bed topography impact simulations of the Antarctic Ice Sheet evolution and how these effects vary across continental-, regional- and glacier-scale simulations. While ice mass loss and dynamic evolution remain qualitatively similar for all the simulations, continental-, regional- and glacier-scale simulations can differ in exact grounding line location and timing of retreat. As simulations are performed with increased resolution, they can both underestimate or overestimate mass loss on decadal and centennial time scales.

Glacier-scale simulations, whose highest resolution is increased to 500 m, display results closer to regional-scale, especially in the Bellingshausen Sea sector. Our results, therefore, suggest that regional-scale simulations are a good scale to capture rapid changes of the Antarctic Ice Sheet on decadal to centennial time scales, similar to what has been shown for Greenland [32], but also that different regions have different behaviours (e.g. depending on the underlying topography, width of the ice streams) and should be assessed individually [51–53].

We combine the Totten and Moscow basins in our glacier-scale simulations, since simulations treating them as separate glaciers (electronic supplementary material, table S1) exhibit unphysical behaviour. The grounding line at their shared boundary retreats substantially under the high-emission scenario. However, glacier-scale simulations separating Totten and Moscow show only limited retreat in this area, leaving a grounding line that is too advanced and a narrow band of grounded ice after 2200 compared with both the regional-scale and the combined Totten–Moscow glacier-scale simulations (electronic supplementary material, fig. S2). This artefact is caused by the grounding-line retreat along the boundary between the two glacier-scale domains and the way boundary conditions are imposed on this shared boundary. A similar, but less extreme, case appears in figure 5c at the Totten–Vincennes boundary after 2200, when the ice shelves of the two glaciers merge. These findings highlight the limitations of glacier-scale simulations over long time scales, as boundary conditions prescribed on their boundaries start to be impacted by large changes.

Additional regional-scale experiments were performed at resolutions comparable to those of the continental- and glacier-scale, with minimum resolutions of 3 km and 500 m, respectively (electronic supplementary material, table S1). Simulations with matching resolutions produced nearly identical results (electronic supplementary material, fig. S4), indicating that model resolution is the primary factor driving differences across scales. Small discrepancies, which tend to amplify over time, are sometimes observed at the same resolution. They are caused by the coarser resolution of continental-scale simulations at basin divides, interactions between glaciers through

ice shelves and the retreat of ice fronts along boundaries of the domain, similar to those described for the glacier-scale simulations above.

(b) Amplitude of bed uncertainties versus other sources of uncertainties

Despite decades of progress in collecting bedrock elevation data and refining methods to process and integrate them, large uncertainties in bed topography remain across many regions beneath the Antarctic Ice Sheet (figure 2). Using uncertainty estimates from BedMachine Antarctica v.4 [27,29], simulations with varying bedrock topographies highlight how bed uncertainty affects projections of the ice sheet's future evolution. Continental-scale simulations suggest that, on century time scales, bed topography uncertainties can exceed the differences between climate scenarios in terms of projected ice loss for the extreme case where the uncertainty of BedMachine Antarctica is applied over the entire ice sheet (see also §4c). Regional simulations reveal even greater sensitivity to bed elevation uncertainties owing to their higher spatial resolution, which enables them to resolve finer-scale topographic features [31,54]. Glacial-scale results are very similar to regional-scale ones for the Bellingshausen Sea region, but are even more sensitive to variations in bed topography for the Aurora region. The impact of bedrock elevation uncertainty is comparable to that introduced by varying physical processes or model parameters [19,20,53], but remains smaller than the spread from large multi-model ensembles [7].

This uncertainty could be even higher for simulations with different parameter choices, such as subglacial hydrology [16], nonlinearity of the friction law and higher melt forcing [55], as bed topography and the nature of the bed have a direct impact on ice sheet dynamics.

(c) Robustness of results and limitations

One major challenge in assessing the sensitivity of ice sheet models to bedrock uncertainty lies in its fundamental influence on model initialization. Since each bed topography affects the model geometry, we must independently parameterize and initialize simulations for each topography, requiring separate inversions to match the observed velocity field. To evaluate the impact of performing multiple inversions, which result in slightly different modelled initial velocities, we conducted two additional continental-scale simulations using the minimum bedrock elevation. In one simulation, we followed the same inversion procedure but used different initial guesses for the friction and rheology fields. Instead of using generic initial guesses, i.e. assuming that the driving stress is exactly compensated by the basal shear stress for friction and that the thermal viscosity is calculated assuming a temperature of -10°C [56], we start from the adjusted fields calculated for the standard case. In the other case, we used the exact same friction and rheology values as in the standard bed case. The grey shaded area in figure 4a shows the range of ice mass change across these three simulations. It confirms that differences in the initial velocities are small enough to have only a limited effect on ice sheet evolution, significantly smaller than the impact of using different bedrock topographies. The results with the three bed topographies, therefore, reflect the impact of the topography itself and not differences from small variations in the initial model velocities.

The results presented here are based on extreme experiments, created by adding or subtracting the BedMachine Antarctica errors beneath the entire Antarctic Ice Sheet. Future work should explore localized variations in bed topography that cover the entire range of possible bedrock topography instead of using only extreme cases to better understand the impact of uncertainty under more realistic conditions, as the presence of pinning points or a retrograde slope can strongly affect grounding line dynamics [57–60]. A study that explores more complex bed topography by performing ensembles of simulations with random topographic noise at the continental scale [61] also suggests that the Aurora region and Filchner–Ronne regions are most affected by bedrock uncertainties. However, in their experiments, the retreat of the grounding line in the Ross and Amundsen Sectors is less sensitive to the bed, as a large retreat is observed for all the experiments

regardless of the bed selected. Such experiments have been carried out for climate conditions typical of the Pliocene and with a much coarser model, which is likely to impact results, as model resolution plays a key role in the impact of bed uncertainty (see §4a,b). Other regional-scale experiments tested the impact of adding noise to the bed topography with a similar amplitude but different frequency [62]. The results suggest that noise with a broad spatial scale of more than tens of kilometres has more impact than higher-frequency noise, even for an identical total noise power. Other results suggest that the exact location and amplitude of the bedrock perturbation are more significant than its shape [31], making it difficult to accurately assess the impact of the bed on the ice sheet dynamics and suggesting that the acquisition of new bed observations is required to limit this source of uncertainty.

The bed topography is also fixed during all our simulations. Nonetheless, the long-term evolution of Antarctica is affected by the bedrock uplift caused by large mass loss. While the uplift is expected to be low in East Antarctica by 2100, this uplift may reach several metres in 2100 in the Amundsen Sea sector and delay the retreat of the grounding line in this region [63,64].

While we assess here the role of uncertainties in bed topography under the Antarctic Ice Sheet, similar uncertainties exist in the ocean bathymetry. This uncertainty affects the ocean circulation on the continental shelf as well as in ice shelf cavities and impacts the transport and availability of modified circumpolar deep water close to the Antarctic ice shelves through bathymetry-constrained oceanic gateways [65–67].

Large uncertainties remain in the future evolution of the Antarctic Ice Sheet [10], as many physical processes, model parameters and climate forcings continue to display large uncertainties despite decades of progress [4,12,18–20,68,69]. As demonstrated in our results, bed topography is an additional parameter affecting Antarctic evolution. Its specific impact varies depending not only on the magnitude of the uncertainty but also on dynamic changes that vary in different basins, suggesting that future observational efforts should work closely with models to best design new acquisition campaigns.

5. Conclusions

Our study shows that uncertainties in bed elevation play an important role in affecting Antarctic Ice Sheet projections, often exceeding the role of emission scenarios, particularly until 2150. While continental-scale simulations capture broad patterns of grounding line retreat, regional-scale simulations reveal enhanced sensitivity to local topographic features. In the Aurora and Bellingshausen Sea regions, errors in bed topography from regional- and glacier-scale simulations increase the uncertainty in projected sea-level contribution by 20–40% by 2300, relative to continental-scale simulations. Future studies should focus on spatially targeted perturbations of bed topography, which would offer more nuanced results of how localized uncertainties influence ice sheet evolution.

Data accessibility. The scripts to reproduce results and figures from this paper are available on Zenodo [70]. The ice flow model ISSM is available on GitHub (<https://github.com/ISSMteam/ISSM>), and version v2026.1 used for the simulations is permanently archived on Zenodo [71]. ISMIP6 climate forcing is available at <https://thegehub.org/dataset-listing>. BedMachine Antarctica v. 4 is described in a manuscript in this same issue [29] and is available on NSIDC (<https://nsidc.org/data/nsidc-0756/versions/4>).

Supplementary material is available online [72].

Declaration of AI use. We have not used AI-assisted technologies in creating this article.

Authors' contributions. J.C.: conceptualization, data curation, formal analysis, investigation, methodology, visualization, writing—original draft, writing—review and editing; H.S.: conceptualization, data curation, formal analysis, funding acquisition, investigation, methodology, project administration, resources, software, supervision, validation, writing—original draft, writing—review and editing; S.N.: conceptualization, funding acquisition, investigation, methodology, resources, validation, writing—review and editing.

All authors gave final approval for publication and agreed to be held accountable for the work performed therein.

Conflict of interest declaration. We declare we have no competing interests.

Funding. Authors are funded by grants from the NASA Sea Level Change Team Program (80NSSC24K1532 and 80NSSC21K0322). H.S. acknowledges support from the Novo Nordisk Foundation under the Challenge Programme 2023—Grant number NNF23OC00807040. This research was supported in part by grant NSF-PHY-2309135 to the Kavli Institute for Theoretical Physics (KITP).

References

1. Rignot E, Mouginot J, Scheuchl B, van den Broeke M, van Wessem MJ, Morlighem M. 2019 Four decades of Antarctic Ice sheet mass balance from 1979–2017. *Proc. Natl Acad. Sci. USA* **116**, 1095–1103. (doi:10.1073/pnas.1812883116)
2. Otosaka IN *et al.* 2023 Mass balance of the Greenland and Antarctic ice sheets from 1992 to 2020. *Earth Syst. Sci. Data* **15**, 1597–1616. (doi:10.5194/essd-15-1597-2023)
3. Nowicki SMJ, Payne T, Larour E, Seroussi H, Goelzer H, Lipscomb W, Gregory J, Abe-Ouchi A, Shepherd A. 2016 Ice Sheet Model Intercomparison Project (ISMIP6) contribution to CMIP6. *Geosci. Model Dev.* **9**, 4521–4545. (doi:10.5194/gmd-9-4521-2016)
4. Nowicki S *et al.* 2020 Experimental protocol for sea level projections from ISMIP6 stand-alone ice sheet models. *Cryosphere* **14**, 2331–2368. (doi:10.5194/tc-14-2331-2020)
5. Seroussi H *et al.* 2020 ISMIP6 Antarctica: a multi-model ensemble of the Antarctic ice sheet evolution over the 21st century. *Cryosphere* **14**, 3033–3070. (doi:10.5194/tc-14-3033-2020)
6. Seroussi H *et al.* 2023 Insights into the vulnerability of Antarctic glaciers from the ISMIP6 ice sheet model ensemble and associated uncertainty. *Cryosphere* **17**, 5197–5217. (doi:10.5194/tc-17-5197-2023)
7. Seroussi H *et al.* 2024 Evolution of the Antarctic ice sheet over the next three centuries from an ISMIP6 Model Ensemble. *Earths. Future* **12**, e2024EF004561. (doi:10.1029/2024EF004561)
8. Holland DM, Nicholls KW, Basinski A. 2020 The Southern Ocean and its interaction with the Antarctic Ice Sheet. *Science* **367**, 1326–1330. (doi:10.1126/science.aaz5491)
9. Smith B *et al.* 2020 Pervasive ice sheet mass loss reflects competing ocean and atmosphere processes. *Science* **368**, 1239–1242. (doi:10.1126/science.aaz5845)
10. Edwards T *et al.* 2021 Quantifying uncertainties in the land ice contribution to sea level rise this century. *Nature* **593**, 74–82. (doi:10.1038/s41586-021-03302-y)
11. Seroussi H *et al.* 2019 initMIP-Antarctica: an ice sheet model initialization experiment of ISMIP6. *Cryosphere* **13**, 1441–1471. (doi:10.5194/tc-13-1441-2019)
12. Nowicki S, Seroussi H. 2018 Projections of future sea level contributions from the Greenland and Antarctic Ice sheets: challenges beyond dynamical ice sheet modeling. *Oceanography* **31**, 109–117. (doi:10.5670/oceanog.2018.216)
13. Bindshadler RA *et al.* 2013 Ice-sheet model sensitivities to environmental forcing and their use in projecting future sea level (the SeaRISE project). *J. Glaciol.* **59**, 195–224. (doi:10.3189/2013JoG12J125)
14. Nowicki S *et al.* 2013 Insights into spatial sensitivities of ice mass response to environmental change from the SeaRISE ice sheet modeling project I: Antarctica. *J. Geophys. Res.* **118**, 1002–1024. (doi:10.1002/jgrf.20081)
15. Kazmierczak E, Sun S, Coulon V, Pattyn F. 2022 Subglacial hydrology modulates basal sliding response of the Antarctic ice sheet to climate forcing. *Cryosphere* **16**, 4537–4552. (doi:10.5194/tc-16-4537-2022)
16. Zhao C *et al.* 2025 Subglacial water amplifies Antarctic contributions to sea-level rise. *Nat. Commun.* **16**, 3187. (doi:10.1038/s41467-025-58375-4)
17. Dawson EJ, Schroeder DM, Chu W, Mantelli E, Seroussi H. 2022 Ice mass loss sensitivity to the Antarctic ice sheet basal thermal state. *Nat. Commun.* **13**, 4957. (doi:10.1038/s41467-022-32632-2)
18. Reese R, Levermann A, Albrecht T, Seroussi H, Winkelmann R. 2020 The role of history and strength of the oceanic forcing in sea level projections from Antarctica with the Parallel Ice Sheet Model. *Cryosphere* **14**, 3097–3110. (doi:10.5194/tc-14-3097-2020)
19. Lipscomb WH, Leguy GR, Jourdain NC, Asay-Davis XS, Seroussi H, Nowicki S. 2021 ISMIP6-based projections of ocean-forced Antarctic Ice Sheet evolution using the Community Ice Sheet Model. *Cryosphere* **15**, 633–661. (doi:10.5194/tc-15-633-2021)

20. Coulon V, Klose AK, Kittel C, Edwards T, Turner F, Winkelmann R, Pattyn F. 2024 Disentangling the drivers of future Antarctic ice loss with a historically calibrated ice-sheet model. *Cryosphere* **18**, 653–681. (doi:10.5194/tc-18-653-2024)
21. McCormack FS, Warner RC, Seroussi H, Dow CF, Roberts JL, Treverrow A. 2022 Modeling the deformation regime of Thwaites Glacier, West Antarctica, using a simple flow relation for Ice Anisotropy (ESTAR). *J. Geophys. Res.* **127**, e2021JF006332. (doi:10.1029/2021JF006332)
22. Brondex J, Gagliardini O, Gillet-chaulet F, Durand G. 2017 Sensitivity of grounding line dynamics to the choice of the friction law. *J. Glaciol.* **63**, 854–866. (doi:10.1017/jog.2017.51)
23. Yu H, Rignot E, Seroussi H, Morlighem M, Choi Y. 2019 Impact of Iceberg calving on the retreat of Thwaites Glacier, West Antarctica over the next century with different calving laws and Ocean Thermal Forcing. *Geophys. Res. Lett.* **46**, 14539–14547. (doi:10.1029/2019GL084066)
24. Bradley AT, Bett DT, Dutrieux P, De Rydt J, Holland PR. 2022 The influence of pine Island Ice shelf calving on basal melting. *J. Geophys. Res.* **127**, e2022JC018621. (doi:10.1029/2022JC018621)
25. Holland PR, Bevan SL, Luckman AJ. 2023 Strong ocean melting feedback during the recent retreat of Thwaites Glacier. *Geophys. Res. Lett.* **50**, e2023GL103088. (doi:10.1029/2023GL103088)
26. Robel AA, Seroussi H, Roe GH. 2019 Marine ice sheet instability amplifies and skews uncertainty in projections of future sea-level rise. *Proc. Natl Acad. Sci. USA* **116**, 14887–14892. (doi:10.1073/pnas.1904822116)
27. Morlighem M *et al.* 2020 Deep glacial troughs and stabilizing ridges unveiled beneath the margins of the Antarctic ice sheet. *Nat. Geosci.* **13**, 132–137. (doi:10.1038/s41561-019-0510-8)
28. Fretwell P *et al.* 2013 Bedmap2: improved ice bed, surface and thickness datasets for Antarctica. *Cryosphere* **7**, 375–393. (doi:10.5194/tc-7-375-2013)
29. Morlighem M, Charrassin R, Maffezzoli N, Millan R, Ockenden H, Schlegel NJ, Seroussi H, Wood M. 2026 New high-resolution subglacial bed topography of the Antarctic and Greenland Ice Sheets. *Phil. Trans. R. Soc. A* **384**, 20240538. (doi:10.1098/rsta.2024.0538)
30. Durand G, Gagliardini O, Favier L, Zwinger T, le Meur E. 2011 Impact of bedrock description on modeling ice sheet dynamics. *Geophys. Res. Lett.* **38**, L20501. (doi:10.1029/2011GL048892)
31. Castleman BA, Schlegel NJ, Caron L, Larour E, Khazendar A. 2022 Derivation of bedrock topography measurement requirements for the reduction of uncertainty in ice-sheet model projections of Thwaites Glacier. *Cryosphere* **16**, 761–778. (doi:10.5194/tc-16-761-2022)
32. Badgeley JA, Morlighem M, Seroussi H. 2025 Increased sea-level contribution from north-western Greenland for models that reproduce observations. *Proc. Natl Acad. Sci. USA* **122**, e2411904122. (doi:10.1073/pnas.2411904122)
33. Larour E, Seroussi H, Morlighem M, Rignot E. 2012 Continental scale, high order, high spatial resolution, ice sheet modeling using the Ice Sheet System Model (ISSM). *J. Geophys. Res.* **117**, 1–20. (doi:10.1029/2011JF002140)
34. Seroussi H, Morlighem M, Larour E, Rignot E, Khazendar A. 2014 Hydrostatic grounding line parameterization in ice sheet models. *Cryosphere* **8**, 2075–2087. (doi:10.5194/tc-8-2075-2014)
35. Seroussi H, Morlighem M. 2018 Representation of basal melting at the grounding line in ice flow models. *Cryosphere* **12**, 3085–3096. (doi:10.5194/tc-12-3085-2018)
36. MacAyeal DR. 1989 Large-scale ice flow over a viscous basal sediment: theory and application to ice stream B, Antarctica. *J. Geophys. Res.* **94**, 4071–4087. (doi:10.1029/JB094iB04p04071)
37. Glen JW. 1955 The creep of polycrystalline ice. *Proc. R. Soc. Lond. A* **228**, 519–538. (doi:10.1098/rspa.1955.0066)
38. Budd WF, Keage PL, Blundy NA. 1979 Empirical studies of ice sliding. *J. Glaciol.* **23**, 157–170. (doi:10.3189/S0022143000029804)
39. MacAyeal DR. 1992 Irregular oscillations of the West Antarctic ice sheet. *Nature* **359**, 29–32. (doi:10.1038/359029a0)
40. Mouginot J, Rignot E, Scheuchl B, Millan R. 2017 Comprehensive annual ice sheet velocity mapping using Landsat-8, Sentinel-1, and RADARSAT-2 data. *Remote Sens.* **9**, 364. (doi:10.3390/rs9040364)
41. Seroussi H, Morlighem M, Rignot E, Khazendar A, Larour E, Mouginot J. 2013 Dependence of century-scale projections of the Greenland ice sheet on its thermal regime. *J. Glaciol.* **59**, 1024–1034. (doi:10.3189/2013JoG13J054)
42. Jourdain NC, Asay-Davis X, Hattermann T, Straneo F, Seroussi H, Little CM, Nowicki S. 2020 A protocol for calculating basal melt rates in the ISMIP6 Antarctic ice sheet projections. *Cryosphere* **14**, 3111–3134. (doi:10.5194/tc-14-3111-2020)

43. Bondzio JH, Seroussi H, Morlighem M, Kleiner T, Rückamp M, Humbert A, Larour EY. 2016 Modelling calving front dynamics using a level-set method: application to Jakobshavn Isbræ, West Greenland. *Cryosphere* **10**, 497–510. (doi:10.5194/tc-10-497-2016)
44. Shepherd A *et al.* 2018 Mass balance of the Antarctic Ice Sheet from 1992 to 2017. *Nature* **558**, 219–222. (doi:10.1038/s41586-018-0179-y)
45. Seroussi H, Morlighem M, Rignot E, Larour E, Aubry D, Ben Dhia H, Kristensen SS. 2011 Ice flux divergence anomalies on 79north Glacier, Greenland. *Geophys. Res. Lett.* **38**, 1–5. (doi:10.1029/2011GL047338)
46. van Wessem JM *et al.* 2018 Modelling the climate and surface mass balance of polar ice sheets using RACMO2 – Part 2: Antarctica (1979–2016). *Cryosphere* **12**, 1479–1498. (doi:10.5194/tc-12-1479-2018)
47. Morlighem M, Rignot E, Seroussi H, Larour E, Ben Dhia H, Aubry D. 2011 A mass conservation approach for mapping glacier ice thickness. *Geophys. Res. Lett.* **38**, 1–6. (doi:10.1029/2011GL048659)
48. Ockenden H, Bingham RG, Curtis A, Goldberg D. 2023 Ice-flow perturbation analysis: a method to estimate ice-sheet bed topography and conditions from surface datasets. *J. Glaciol.* **69**, 1677–1686. (doi:10.1017/jog.2023.50)
49. Mouginot J, Rignot E, Scheuchl B. 2019 Continent-wide, interferometric SAR phase, mapping of Antarctic ice velocity. *Geophys. Res. Lett.* **46**, 9710–9718. (doi:10.1029/2019GL083826)
50. Sellar AA *et al.* 2019 UKESM1: description and evaluation of the U.K. Earth system model. *J. Adv. Model. Earth Syst.* **11**, 4513–4558. (doi:10.1029/2019MS001739)
51. Gollidge NR, Levy RH, McKay RM, Naish TR. 2017 East Antarctic ice sheet most vulnerable to Weddell Sea warming. *Geophys. Res. Lett.* **44**, 2343–2351. (doi:10.1002/2016GL072422)
52. Bulthuis K, Arnst M, Sun S, Pattyn F. 2019 Uncertainty quantification of the multi-centennial response of the Antarctic ice sheet to climate change. *Cryosphere* **13**, 1349–1380. (doi:10.5194/tc-13-1349-2019)
53. Schlegel NJ, Seroussi H, Schodlok MP, Larour EY, Boening C, Limonadi D, Watkins MM, Morlighem M, van den Broeke MR. 2018 Exploration of Antarctic Ice Sheet 100-year contribution to sea level rise and associated model uncertainties using the ISSM framework. *Cryosphere* **12**, 3511–3534. (doi:10.5194/tc-12-3511-2018)
54. Williams CR, Thodoroff P, Arthern RJ, Byrnes J, Hosking JS, Kaiser M, Lawrence ND, Kazlauskaitė I. 2025 Calculations of extreme sea level rise scenarios are strongly dependent on ice sheet model resolution. *Commun. Earth Environ* **6**, 60. (doi:10.1038/s43247-025-02010-z)
55. Wernecke A, Edwards TL, Holden PB, Edwards NR, Cornford SL. 2022 Quantifying the impact of bedrock topography uncertainty in Pine Island glacier projections for this century. *Geophys. Res. Lett.* **49**, s2021GL096589. (doi:10.1029/2021GL096589)
56. Cuffey KM, Paterson WSB. 2010 *The physics of glaciers*, 4th edn. Oxford, UK: Elsevier.
57. Favier L, Gagliardini O, Durand G, Zwinger T. 2012 A three-dimensional full stokes model of the grounding line dynamics: effect of a pinning point beneath the ice shelf. *Cryosphere* **6**, 101–112. (doi:10.5194/tc-6-101-2012)
58. Gudmundsson GH, Krug J, Durand G, Favier L, Gagliardini O. 2012 The stability of grounding lines on retrograde slopes. *Cryosphere* **6**, 1497–1505. (doi:10.5194/tc-6-1497-2012)
59. Robel AA, Schoof C, Tziperman E. 2016 Persistence and variability of ice-stream grounding lines on retrograde bed slopes. *Cryosphere* **10**, 1883–1896. (doi:10.5194/tc-10-1883-2016)
60. Wild CT, Alley KE, Muto A, Truffer M, Scambos TA, Pettit EC. 2022 Weakening of the pinning point buttressing Thwaites Glacier, West Antarctica. *The Cryosphere* **16**, 397–417. (doi:10.5194/tc-16-397-2022)
61. Gasson E, DeConto R, Pollard D. 2015 Antarctic bedrock topography uncertainty and ice sheet stability. *Geophys. Res. Lett.* **42**, 5372–5377. (doi:10.1002/2015GL064322)
62. Sun S, Cornford SL, Liu Y, Moore JC. 2014 Dynamic response of Antarctic ice shelves to bedrock uncertainty. *Cryosphere* **8**, 1561–1576. (doi:10.5194/tc-8-1561-2014)
63. Adhikari S, Ivins ER, Larour E, Seroussi H, Morlighem M, Nowicki S. 2014 Future Antarctic bed topography and its implications for ice sheet dynamics. *Solid Earth* **5**, 569–584. (doi:10.5194/se-5-569-2014)
64. Larour E, Seroussi H, Adhikari S, Ivins E, Caron L, Morlighem M, Schlegel N. 2019 Slowdown in Antarctic mass loss from solid Earth and sea-level feedbacks. *Science* **364**, eaav7908. (doi:10.1126/science.aav7908)

65. Nakayama Y, Timmermann R, Schröder M, Hellmer HH. 2014 On the difficulty of modeling circumpolar deep water intrusions onto the Amundsen Sea continental shelf. *Ocean Model* **84**, 26–34. (doi:[10.1016/j.ocemod.2014.09.007](https://doi.org/10.1016/j.ocemod.2014.09.007))
66. de Boer AM, Hutchinson DK, Roquet F, Sime LC, Burls NJ, Heuzé C. 2022 The Impact of Southern Ocean Topographic Barriers on the Ocean Circulation and the overlying Atmosphere. *J. Clim.* **35**, 5805–5821. (doi:[10.1175/JCLI-D-21-0896.1](https://doi.org/10.1175/JCLI-D-21-0896.1))
67. Nicola L, Reese R, Kreuzer M, Albrecht T, Winkelmann R. 2025 Bathymetry-constrained warm-mode melt estimates derived from analysing oceanic gateways in Antarctica. *Cryosphere* **19**, 2263–2287. (doi:[10.5194/tc-19-2263-2025](https://doi.org/10.5194/tc-19-2263-2025))
68. Pattyn F, Favier L, Sun S, Durand G. 2017 Progress in numerical modeling of Antarctic Ice-Sheet dynamics. *Curr. Clim. Change Rep.* **3**, 174–184. (doi:[10.1007/s40641-017-0069-7](https://doi.org/10.1007/s40641-017-0069-7))
69. Goelzer H, Robinson A, Seroussi H, van de Wal RSW. 2017 Recent progress in Greenland Ice sheet modelling. *Curr. Clim. Change Rep.* **3**, 291–302. (doi:[10.1007/s40641-017-0073-y](https://doi.org/10.1007/s40641-017-0073-y))
70. Seroussi H, Caillet J, Nowicki S. 2025 Data and code for manuscript: Uncertain ground: Impact of bed topography on Antarctic Ice Sheet projections. Zenodo (doi:[10.5281/zenodo.17237334](https://doi.org/10.5281/zenodo.17237334))
71. Morlighem M *et al.* 2026 ISSMteam/ISSM: 2026.1 (2026.1). Zenodo. (doi:[10.5281/zenodo.18166185](https://doi.org/10.5281/zenodo.18166185))
72. Caillet J, Seroussi H, Nowicki S. 2026 Supplementary material from: Uncertain ground: Impact of bed topography on Antarctic ice sheet projections. Figshare. (doi:[10.6084/m9.figshare.c.8405335](https://doi.org/10.6084/m9.figshare.c.8405335))